Orthogonal trajectories.

Step 1: find $\frac{dy}{dx}$ & c, put value of c in $\frac{dy}{dx}$

Step 2: If eqn can be solved with no V, just do it!

Ex: 1, 2, 3, 4 | Ex: 5: needs $V = \frac{y}{x}$

Step 3: solve eqn.

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1) $y^2 = cx$ 2) $cx^2 + y^2 = 1$ 3) $y = \frac{cx^2}{x+1}$ ~~4) y = ce~~ 4) $y = e^{cx}$

5) $x^2 + y^2 = cx^3$

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(2) Given,

$$cx^2 + y^2 = 1$$

$$\Rightarrow y^2 = 1 - cx^2$$

$$\Rightarrow 2y\frac{dy}{dx} = -2xc.$$

$$\Rightarrow y\frac{dy}{dx} = -x \cdot \frac{1-y^2}{x^2} \qquad \left[\because c = \frac{1-y^2}{x^2}\right]$$

$$\Rightarrow y\frac{dy}{dx} = \frac{y^2-1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2-1}{xy}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-xy}{y^2-1}$$

$$\Rightarrow (y^2-1)\,dy = (-xy)\,dx$$

$$\Rightarrow \left(\frac{y^2-1}{y}\right)dy = -x\,dx$$

$$\Rightarrow \int \frac{y^2}{y}\,dy - \int \frac{1}{y}\,dy = -\int x\,dx$$

$$\Rightarrow \frac{y^2}{2} - \ln y = -\frac{x^2}{2}$$

$$\Rightarrow \ln y - \frac{y^2}{2} = \frac{x^2}{2} + c$$

$$\Rightarrow \ln y - \frac{y^2}{2} - \frac{x^2}{2} = c$$

(Ans.)

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(3) Given,

$$y = \frac{cx^2}{x+1}$$

$$\Rightarrow \frac{dy}{dx} = c\,\frac{d}{dx}\left(\frac{x^2}{x+1}\right)$$

$$\Rightarrow \frac{dy}{dx} = c\,\frac{(x+1)\cdot 2x - x^2\cdot 1}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(x+1)}{x^2} \times \frac{(x+1)\cdot 2x - x^2}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(2x^2+2x) - x^2y}{x^2(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x(2xy+2y-xy)}{x^2(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(2x+2-x)}{x(x+1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(x+2)}{x(x+1)}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal

$$\therefore \frac{dy}{dx} = \frac{-x(x+1)}{y(x+2)}$$

$$\Rightarrow y(x+2)\,dy = -x(x+1)\,dx$$

$$\Rightarrow y\,dy = \left(\frac{-x^2-x}{x+2}\right)dx$$

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$$\Rightarrow \int y\, dy = \int \frac{-x^2 - x + 2 - 2}{x+2} dx$$

$$\Rightarrow \int y\, dy = \int \frac{-x^2 - 2x + x + 2 - 2}{x+2} dx$$

$$\Rightarrow \int y\, dy = \int \frac{-x(x+2) + 1(x+2) - 2}{(x+2)} dx$$

$$\Rightarrow \int y\, dy = \int \left\{ \frac{(x+2)(1-x)}{x+2} - \frac{2}{x+2} \right\} dx$$

$$\Rightarrow \int y\, dy = \int \frac{(x+2)(1-x)}{x+2} dx - 2\int \frac{1}{x+2} dx$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} - 2\ln(x+2) + c$$

$$\Rightarrow \frac{y^2}{2} + \frac{x^2}{2} + 2\ln(x+2) = c$$

(Ans.)

5) Given,

$$x^2 + y^2 = cx^3 \quad \cdots\cdots \text{(1)}$$

$$\therefore 2x + 2y \frac{dy}{dx} = 3cx^2$$

$$\Rightarrow 2y \frac{dy}{dx} = 3cx^2 - 2x$$

$$\Rightarrow 2y \frac{dy}{dx} = 3\left(\frac{x^2 + y^2}{x^3}\right) \cdot x^2 - 2x$$

$$\Rightarrow 2y \frac{dy}{dx} = \frac{3x^2 + 3y^2}{x} - 2x$$

$$\Rightarrow 2y\frac{dy}{dx} = \frac{3x^2+3y^2-2x^2}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2+3y^2}{2xy}$$

Replace $\frac{dy}{dx}$ by it's negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-2xy}{x^2+3y^2} \quad \text{---} ②$$

Let,

$$v = \frac{y}{x}$$

$$\Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$$

put $\frac{dy}{dx}$ and y in ②,

$$v + x\frac{dv}{dx} = \frac{-2x\cdot vx}{x^2+3(vx)^2}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-2vx^2}{x^2+3v^2x^2} - v$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-2v}{1+3v^2} - v$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-2v-v-3v^3}{1+3v^2}$$

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$$\Rightarrow x\frac{dv}{dx} = \frac{-3v^3-3v}{1+3v^2}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-3(v^3+v)}{1+3v^2}$$

$$\Rightarrow \frac{1+3v^2}{v^3+v}\,dv = \frac{-3}{x}\,dx$$

$$\Rightarrow \int\frac{1+3v^2}{v^3+v}\,dv + \int\frac{3}{x}\,dx = 0$$

$$\Rightarrow \ln(v^3+v) + 3\ln x = \ln c$$

$$\Rightarrow \ln(v^3+v) + \ln x^3 = \ln c$$

$$\Rightarrow \ln\{x^3\cdot(v^3+v)\} = \ln c$$

$$\Rightarrow x^3(v^3+v) = c$$

$$\Rightarrow x^3\left(\frac{y^3}{x^3}+\frac{y}{x}\right) = c$$

$$\Rightarrow x^3\times\frac{y^3}{x^3} + x^3\frac{y}{x} = c$$

$$\Rightarrow y^3 + x^2y = c$$

(Ans.)

4) Given,

$$y = e^{cx}$$

$$\Rightarrow \ln y = cx$$

$$\Rightarrow c = \frac{\ln y}{x}$$

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$\therefore y = e^{cx}$

$\Rightarrow \frac{dy}{dx} = ce^{cx}$

$\Rightarrow \frac{dy}{dx} = \frac{\ln y}{x} \cdot y \qquad \left[\because c = \frac{\ln y}{x}\ ;\ y = e^{cx}\right]$

$\Rightarrow \frac{dy}{dx} = \frac{y \ln y}{x}$

Replace $\frac{dy}{dx}$ by it's negative reciprocal,

$\therefore \frac{dy}{dx} = \frac{-x}{y \ln y}$

$\Rightarrow y \ln y \, dy = -x \, dx$

$\Rightarrow \int y \ln y \, dy + \int x \, dx = 0$

$\Rightarrow \ln y \int y \, dy - \int \left[\frac{d}{dx}(\ln y) \int y \, dy\right] dy + \frac{x^2}{2} = 0$

$\Rightarrow \ln y \cdot \frac{y^2}{2} - \int \left(\frac{1}{y} \cdot \frac{y^2}{2}\right) dy + \frac{x^2}{2} = c$

$\Rightarrow \ln y \cdot \frac{y^2}{2} - \frac{1}{2} \times \frac{y^2}{2} + \frac{x^2}{2} = c$

$\Rightarrow \frac{y^2 \ln y}{2} - \frac{y^2}{4} + \frac{x^2}{2} = c$

(Ans)

Given,

$$y^2 = cx$$

$$\therefore 2y\frac{dy}{dx} = c$$

$$\Rightarrow 2y\frac{dy}{dx} = \frac{y^2}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{2x}$$

Replace $\frac{dy}{dx}$ by its negative reciprocal,

$$\therefore \frac{dy}{dx} = \frac{-2x}{y}$$

$$\Rightarrow y\,dy = -2x\,dx$$

$$\Rightarrow \int y\,dy = -2\int x\,dx$$

$$\Rightarrow \frac{y^2}{2} = -2\cdot\frac{x^2}{2} + c$$

$$\Rightarrow \frac{y^2}{2} = -x^2 + c$$

$$\Rightarrow y^2 = -2x^2 + c$$

$$\Rightarrow y^2 + 2x^2 = c$$

(Ans.)